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FA23-BCE-120

$\hat{y} = mx + b_0$

Learning rate
$\hookrightarrow \lambda = 0.01635$

$m_0 = 3.7426, \; b_0 = 54.8732$

x	y	$\hat{y}$	$y-\hat{y}$	$x[y-\hat{y}]$	$\hat{y}$ iter 2	$y-\hat{y}$	$x[y-\hat{y}]$
2	69	62.3584	6.6416	13.2832	64.358	4.642	9.284
9	98	88.5566	9.4434	84.9906	97.556	0.444	3.996
5	82	73.5862	8.4138	~~68.9316~~ 42.069	78.585	3.415	17.075
5	77	73.5862	3.4138	17.069	78.585	-1.585	-7.925
3	71	66.101	4.899	14.697	69.101	1.899	5.697
7	84	81.0714	2.9286	20.5002	88.071	-4.071	-28.497
1	55	58.6158	-3.6158	-3.6158	59.616	-4.616	-4.616
8	94	84.814	9.186	73.488	92.813	1.187	9.496
6	84	77.328	6.6712	40.0272	83.328	0.672	4.032
2	64	62.358	1.6416	3.2832	64.358	-0.358	-0.716
$\sum_{i=1}^{n}(y-\hat{y})$			49.6232	$\sum_{i=1}^{n} x(y-\hat{y}) = 305.7916$		1.629	~~1.62~~, 7.826

iter 1

$$\text{grad } m = \frac{-2}{n} \times \sum_{i=1}^{n} x(y-\hat{y}) = \frac{-2}{10}(305.7916)$$

$$\boxed{\text{grad } m = -61.15832}$$

$$\text{grad } b = \frac{-2}{n} \times \sum (y-\hat{y}) = -\frac{1}{5}(49.6232)$$

$$\boxed{\text{grad } b = -9.92464}$$

$$m_1 = 3.7426 - 0.01635(-61.15832) = 4.742538$$

$$b_1 = 54.8732 - 0.01635(-9.92464) = 55.035468$$

iter 2

$$\text{grad } m = -\frac{1}{5}(\cancel{1.629}\ 7.826) = \boxed{-1.5652}$$

$$\text{grad } b = -\frac{1}{5}(1.629) = \boxed{-0.3258}$$

$$m_2 = 4.7425 - 0.01635(-1.5652) = \boxed{4.7681}$$

$$b_2 = 55.035 - 0.01635(-0.3258) = \boxed{55.0403}$$

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